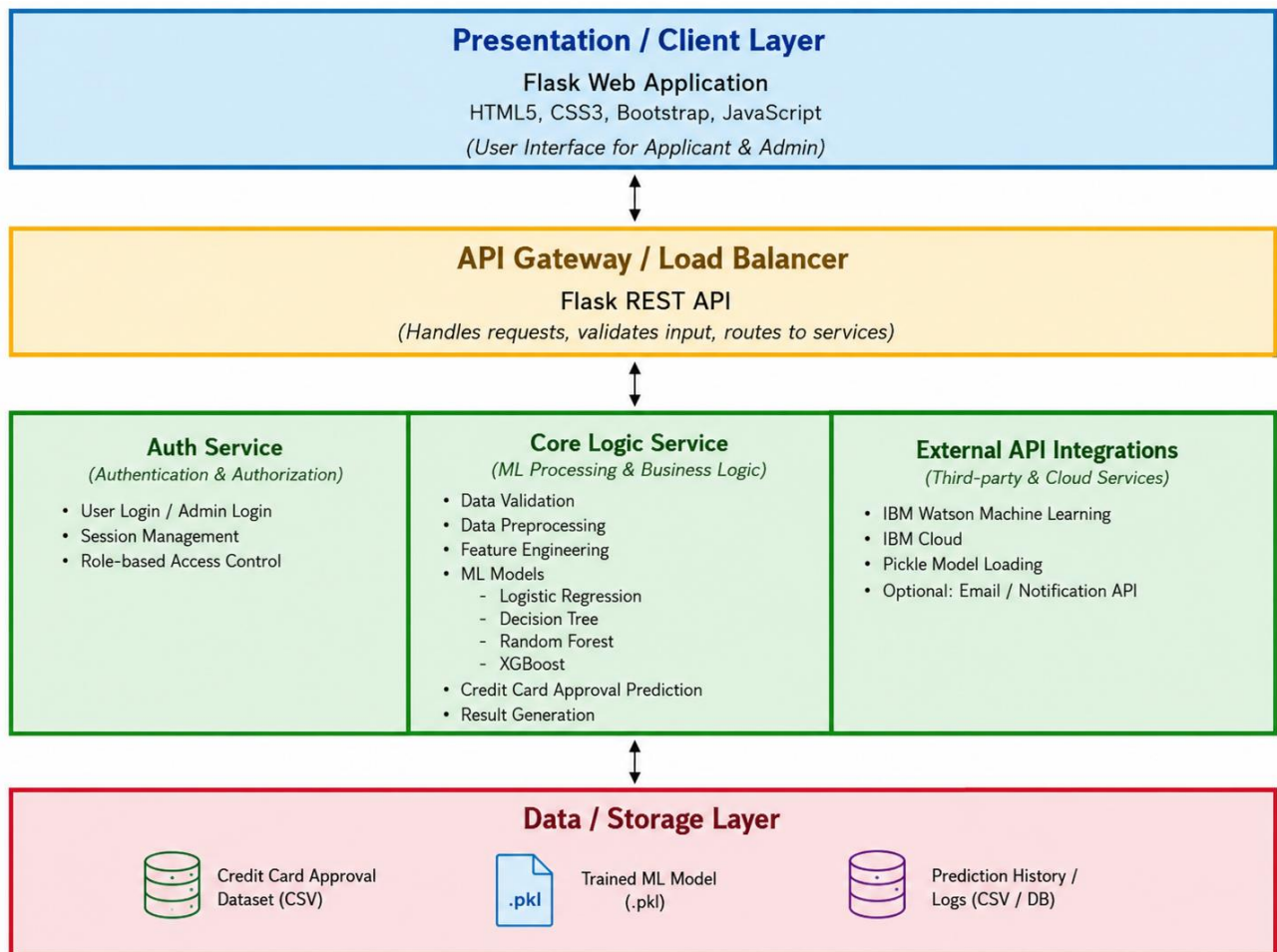


Solution Architecture

Date	03 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface where applicants enter personal and financial details and view credit card approval predictions.	HTML5, CSS3, Bootstrap, JavaScript, Flask (Frontend Templates)
API Gateway	Receives user requests, validates input data, routes requests to the machine learning model, and returns prediction results to the client.	Flask REST API, Python
Application / Core Logic Services (Microservices)	Performs data validation, preprocessing, feature engineering, loads the trained machine learning model, predicts credit card approval, and integrates with IBM Watson Machine Learning for cloud deployment.	Python, Scikit-learn, XGBoost, Pandas, NumPy, Flask, IBM Watson Machine Learning
Database / Data Storage	Stores the credit card approval dataset, trained machine learning model, and prediction history/logs used by the application.	CSV Dataset, Pickle (.pkl) Model, IBM Cloud Storage (optional)